Hridya Sharma

2nd year student (B.Tech Instrumentation & Control Engineering) | Dr. B.R. Ambedkar NIT Jalandhar

Email: hridyas.ic.25@nitj.ac.in

Phone: 7888939304

LinkedIn | GitHub

Professional Profile

Engineering student with hands-on full-stack and data skills (FastAPI, React, PostgreSQL, NumPy, Pandas) and a growing focus on machine learning. Built and deployed full-stack projects and bring strong communication and teamwork skills from debate and consulting work.

Education

B.Tech, Instrumentation & Control Engineering Dr. B.R. Ambedkar National Institute of Technology, Jalandhar

- CGPA: 8.44 (first year)

Technical Skills

- **Languages:** C, C++, Python, JavaScript
- **Data & ML:** NumPy, Pandas, Matplotlib, Streamlit, Folium, Google Earth Engine, basic OpenCV; scikit-learn and machine learning (currently learning)
- **Web Development:** HTML, CSS, React.js, Node.js, Express.js, Tailwind CSS, REST APIs
- **Backend & Databases:** FastAPI, Django (learning), PostgreSQL, SQLAlchemy, Alembic, MongoDB
- **Tools:** Git & GitHub, Vercel, Render
- **Problem Solving:** 230+ LeetCode problems solved (Data Structures & Algorithms)

Projects

CartVerse: Full-Stack E-Commerce App | FastAPI, PostgreSQL, React, Tailwind

- Built and deployed a full-stack e-commerce app (Vercel, Render, Neon) with JWT authentication, product catalog, cart and reviews with verified-purchase enforcement.
- Implemented atomic checkout using SELECT FOR UPDATE row locking, idempotent mock payments, inventory management and Alembic migrations.

HSNMatch: HSN/SAC Code Classifier | Python, PostgreSQL (pgvector), Claude API

- Matches product descriptions to HSN/SAC codes using vector embeddings, then re-ranks candidates with an LLM; built for a finance use case at 180DC

Urban Heat Island Mitigation Simulator (ISRO BAH 2026 Prototype) | Python, Streamlit, Folium, Google Earth Engine

- Built an interactive geospatial dashboard that maps land surface temperature from Landsat 8 imagery and flags urban heat hotspots across Delhi NCR (Najafgarh, Bawana, Okhla, Palam, Badarpur).
- Computed NDVI from Sentinel-2 imagery to visualize green cover, with switchable map layers.
- Developed an intervention sandbox that estimates temperature reduction from tree canopy, cool roofs and surface albedo using a simplified physics-inspired model.

Personal Expense Analyzer | Python, Pandas, Matplotlib

- Cleans raw expense CSVs, generates category summaries, flags budget overruns and plots spending charts

Experience

Member, 180 Degrees Consulting (180DC), NIT Jalandhar

- Part of the student-run consulting chapter that delivers pro bono strategy and analytics projects for NGOs and social enterprises

Member, Literary and Debating Club (LADC), NIT Jalandhar

- Represented the club at multiple Model United Nations (MUN) conferences.
- Built public speaking, argumentation and quick-thinking skills through regular debating.

Achievements

- Won multiple inter-school and inter-college debate competitions, building strong public speaking and quick thinking under pressure.